Subgroup extraction needs the correct modulus

Result. Lean refutes the unqualified algebraic rule that a nonzero ambient coefficient suffices to extract a discrete logarithm. It proves the corrected rule for every additive commutative group and exact prime-order point, gives unbounded finite cofactor examples, and checks actual elliptic point-group controls. Signed standing unverified. [pcc:unverified]

Actual E/F29: $y^2=x^3+1$, #E=30, P=(4,6), order(P)=5

β=5, α=0	β=2, α=3	β=1, α=29
Nonzero mod30	Not invertible mod30	Unit mod30
Zero mod5: no log info	Invertible mod5: log1	Ambient recovery gives1

Generic repair: reduce the actual relation modulo the exact prime subgroup order.

Source condition. Thériault2003 §§2.1/2.4/2.5 assumes a prime-order D_1 in practice but keeps coefficients and linear algebra modulo the full Jacobian order. Printed p78 then tests only β=0 before extracting $-\alpha/\beta$; §5.10 repeats the ambient modulus. A prime subgroup need not have prime ambient order. The exact printed page was rendered and visually inspected. This packet audits that algebraic inference; it does not claim an executed full higher-genus algorithm counterexample. [pcc:unverified]

Generic repair. For any actual additive group, P of exact prime order r and relation aP+b(xP)=0, the order criterion gives r|a+bx. In the actual field ZMod r, b≠0 therefore implies x=−a/b. Division is in the subgroup modulus. For actual ambient ZMod(2r), P=2 has order r and the ambient cardinality is2r. The coefficient β=r is nonzero modulo2r, nonunit, zero modulo r, and gives the same relation for every candidate x. Euclid supplies examples of unbounded prime order; P≠0 checks genuine distinct candidate logs. [pcc:unverified]

Actual curve controls. Lean defines the nonsingular curve $y^2=x^3+1$ over F29, constructs P=(4,6), proves its exact order5 and counts every curve point via an explicit point/code equivalence, obtaining30. β5 is nonzero modulo30 and yields0P+5P=0, but raw inverse multiplication returns0 rather than the log1 of P. A universal ambient-extraction implication is directly negated. The useful relation3P+2P=0 recovers1 modulo5 although2 is not invertible modulo30. Ambient β1 is a unit and recovers1 as a positive control. No composite division-field instance is fabricated. [pcc:unverified]

Attribution and scope. N.Thériault, Index Calculus Attack for Hyperelliptic Curves of Small Genus, LNCS2894(2003),75–92, DOI10.1007/978-3-540-40061-5_5. Relevant original passages, prior immutable source review/curve proof and pinned Mathlib group-order/modular/point-count APIs read. Full genus≥3 Jacobian/Picard identification and a realized algorithm relation trace remain open. No full-paper read, full-runtime refutation or failure-probability refutation is claimed; the latter also needs a coefficient distribution. Historical corpus/backlog labels remain unchanged. [pcc:unverified]

Assurance. Whole gate 28/0failures/0trusted; complete source binding/custom source/config/artifact pins, failed/successful attempts, SVG/PDF retained. Scalar supplement IDC1hc2ffb42b1e7ce04d has a generic symbolic Lean commutative-ring proof and official32+32-point check; it does not prove curve order or extraction. Only propext/Classical.choice/Quot.sound; no holes/custom axioms/native evaluator/raised limits/global flags/waivers. Lean4.35.0/Mathlib56a00585… pinned. Exact PCC replay and signed standing are separate. KB unavailable, bounded source/formal fallback; no native experiment, paid compute or upload. [pcc:unverified]